Question - 16: Write a Python program to implement a Feed Forward Neural Network.

Program Code

```
import numpy as np

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

inputs = np.array([[0.5, 0.8]])

weights_input_hidden = np.array([
    [0.2, 0.4],
    [0.3, 0.7]
])

bias_hidden = np.array([[0.1, 0.1]])

weights_hidden_output = np.array([
    [0.6],
    [0.9]
])

bias_output = np.array([[0.2]])

hidden_input = np.dot(inputs, weights_input_hidden)
+ bias_hidden
hidden_output = sigmoid(hidden_input)

final_input = np.dot(hidden_output,
weights_hidden_output) + bias_output
```

```
final_output = sigmoid(final_input)

print("Input:", inputs)
print("Hidden Layer Output:", hidden_output)
print("Final Output:", final_output)
```

Program Output

```
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB> & C:/Users/Admin/AppD
/Local/Programs/Python/Python314/python.exe "e:/Desktop/SIMATS/Artificial
elligence/AI LAB/experiment16_feed_forward_neural_network.py"
Input: [[0.5 0.8]]
Hidden Layer Output: [[0.60825903 0.70266065]]
Final Output: [[0.76805303]]
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB>
```